

Conjecture 1079: a sparse CSS family violates the distance–girth bound

Local development pilot; not an upstream submission

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The source asserts the universal inequality $d/g \leq \log(g)/2$ for CSS quantum LDPC codes. We refute this necessary first clause with an explicit family. We use the standard binary Pauli definition of quantum distance and the bipartite Tanner graph of both X and Z checks. The source imposes neither connectedness nor increasing distance. Our family has uniformly bounded check weights and qubit incidences and unbounded block length. Both natural and binary logarithms, and every base at least two, give a contradiction. The claimed further extremality and high-girth clauses need not be separately refuted once this universal clause fails.

Theorem 1. *For every positive integer m there is a binary CSS code on $25m$ qubits whose quantum minimum distance is 5, whose combined Tanner graph has girth 4, whose check rows have weight at most 10, and whose qubit incidences are at most 4. Thus this unbounded-length LDPC family violates the stated bound for natural logarithm and for logarithm to every base $b \geq 2$.*

Proof. First construct one block. Label its 25 qubits by $(i, j) \in \{0, 1, 2, 3, 4\}^2$. For each $a \in \{0, 1, 2, 3\}$, let the X check flip all qubits in blocks $i = a, a + 1$. For each i and $a \in \{0, 1, 2, 3\}$, let the Z check flip just $(i, a), (i, a + 1)$. Every X/Z intersection has size zero or two, so the two binary check matrices are orthogonal. Their pure X/Z Pauli labels span a commuting stabilizer S in $(\mathbb{F}_2^{25})^2$. Write a Pauli label as (x, z) ; its weight is the cardinality of the union of their nonzero coordinates, and the quantum distance is

$$d = \min\{\text{wt}(x, z) : (x, z) \in S^\perp \setminus S\}.$$

The orthogonal complement here uses the actual binary symplectic form $x \cdot z' + z \cdot x'$, not the classical Hamming distance of one check code.

In a five-coordinate binary vector space, the four adjacent-pair vectors span exactly the even-parity vectors, and their dot-product constraints force a vector to be constant. These two finite facts are checked directly in Lean over 32 vectors and 16 coefficient lists. Consequently the X rowspace consists of vectors constant on each block, with an even-parity list of the five block values; the Z rowspace consists of vectors having even parity separately in every block. The full centralizer condition says that x is constant on each block and that the five block parities of z are all equal.

A centralizer label outside S has either a nonzero X block, which supplies five supported qubits, or zero x and odd Z parity in every block, which supplies at least one supported qubit per block. (The same lower bound also covers any nonzero x whose block-value parity is even.) Hence its weight is at least five. The pure X label supported on exactly one whole block commutes with every check and is not in the X rowspace; it attains weight five. This proves the actual quantum distance, including both the minimum lower bound and an attaining logical Pauli.

For arbitrary $m > 0$, take the block diagonal X and Z matrices of m such codes. The stabilizer and centralizer decompose into these actual blocks; these are proved rowspace and symplectic

identities. Any global logical Pauli has a logical component in some block. Its global support has cardinality at least that component's support, so its weight is at least five. Extending the attaining witness by zero in all other blocks attains five, proving $d = 5$ for every family member.

The combined Tanner graph has a check vertex for each row of either type and a qubit vertex for each coordinate, with matrix incidence as adjacency. It is bipartite, so has no three-cycle. In the first block copy, X checks $a = 0$ and $a = 1$ share every qubit of block $i = 1$; choosing two distinct such qubits gives a genuine four-cycle with four distinct vertices. Thus its shortest cycle length is $g = 4$. This uses the actual simple graph, not an assumed girth value or an abstract numerical proxy.

X check weights are ten and Z check weights two. Every qubit belongs to at most two X checks and two Z checks, so combined incidence is at most four. These bounds survive block diagonal replication. The qubit count is $25m$, which is unbounded; thus the matrices form a uniformly sparse LDPC family. Finally every member has

$$\frac{d}{g} = \frac{5}{4} > 1 = \frac{\log_2 4}{2} > \frac{\ln 4}{2}.$$

For $b \geq 2$, monotonicity of logarithm gives $\log_b 4 \leq 2$, so the same contradiction holds. No distance, girth, orthogonality or family-admissibility premise is left as an unproved assumption. $\square$

Lean correspondence. The seven source modules implement the actual binary CSS objects, the two five-bit facts, the 25-qubit code, arbitrary diagonal copies, the Tanner graph, degree bounds and the final contradiction. The generated-stabilizer span identity and full centralizer tests bridge the check matrices to quantum distance. The final family predicate requires uniform row and column bounds and unbounded length. Its universal bound quantifies over every member of every such binary CSS family. The checked final negative theorems cover natural logarithm, binary logarithm and all bases at least two. The supplied correspondence file lists their precise names and quantifiers.

Provenance and limits. This is the known generalized Shor construction, used as a counterexample; we do not claim to have discovered a new quantum code. Earlier Steane-code partial work gave distance three and girth four but did not refute the plausible binary-log convention. That route was stopped after independent review; its partial artifacts remain separately recorded. The source does not specify a log base or exclude disconnected families; these conventions are disclosed for independent review. We claim no result for bases between one and two and no separately formalized encoded-qubit dimension. Local verification is separate from submission eligibility, first-solve status and upstream acceptance.